Weather Data Analysis Report

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Objective

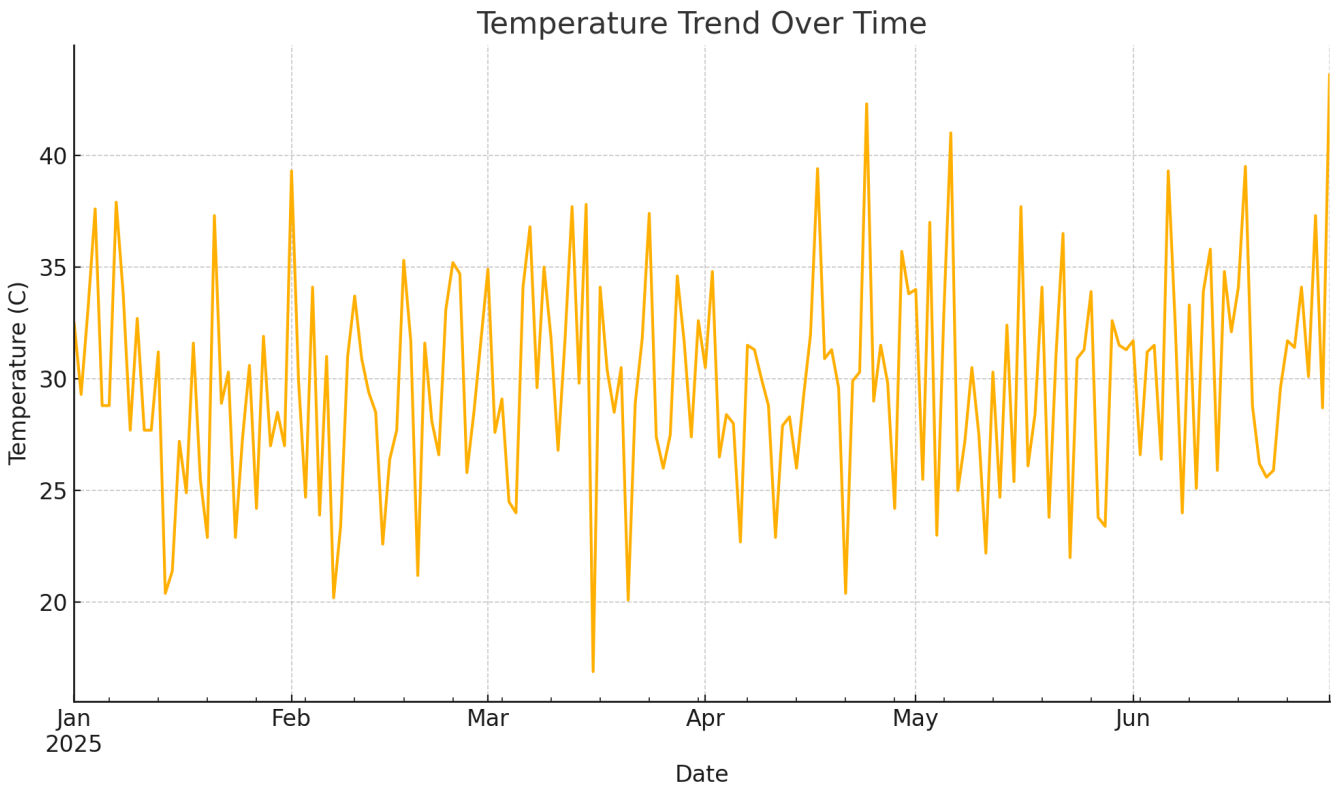
The objective of this project is to perform data preprocessing and exploratory data analysis on a weather dataset. Visual insights are extracted to better understand patterns such as temperature changes, humidity variations, humidity distribution and rainfall behavior over time.

Methodology

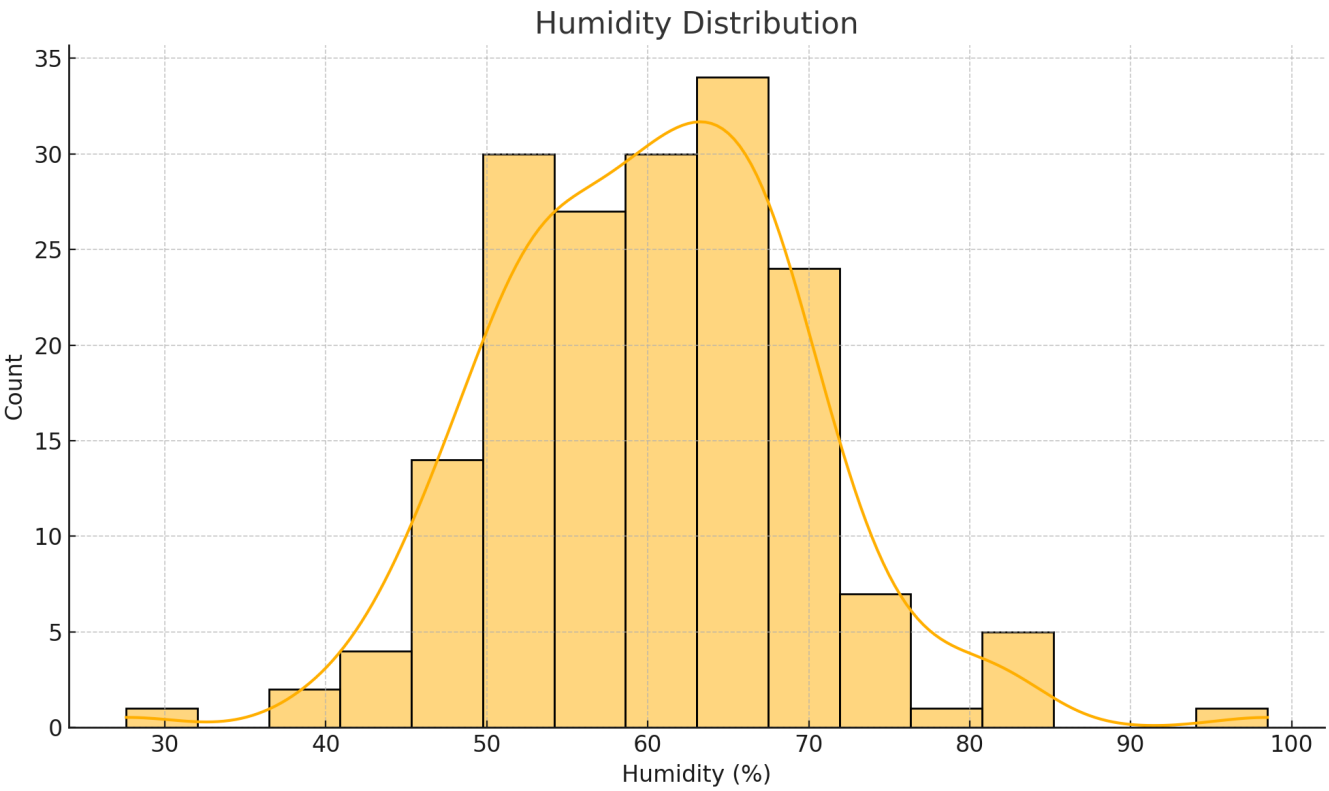
The steps followed in this project are:

1. Dataset creation using Python (synthetic weather data).
2. Basic preprocessing of numerical columns.
3. Setting Date as index for time-series analysis.
4. Exploratory Data Analysis (EDA) using summary statistics.
5. Data visualization using line plot, histogram, boxplot and heatmap.

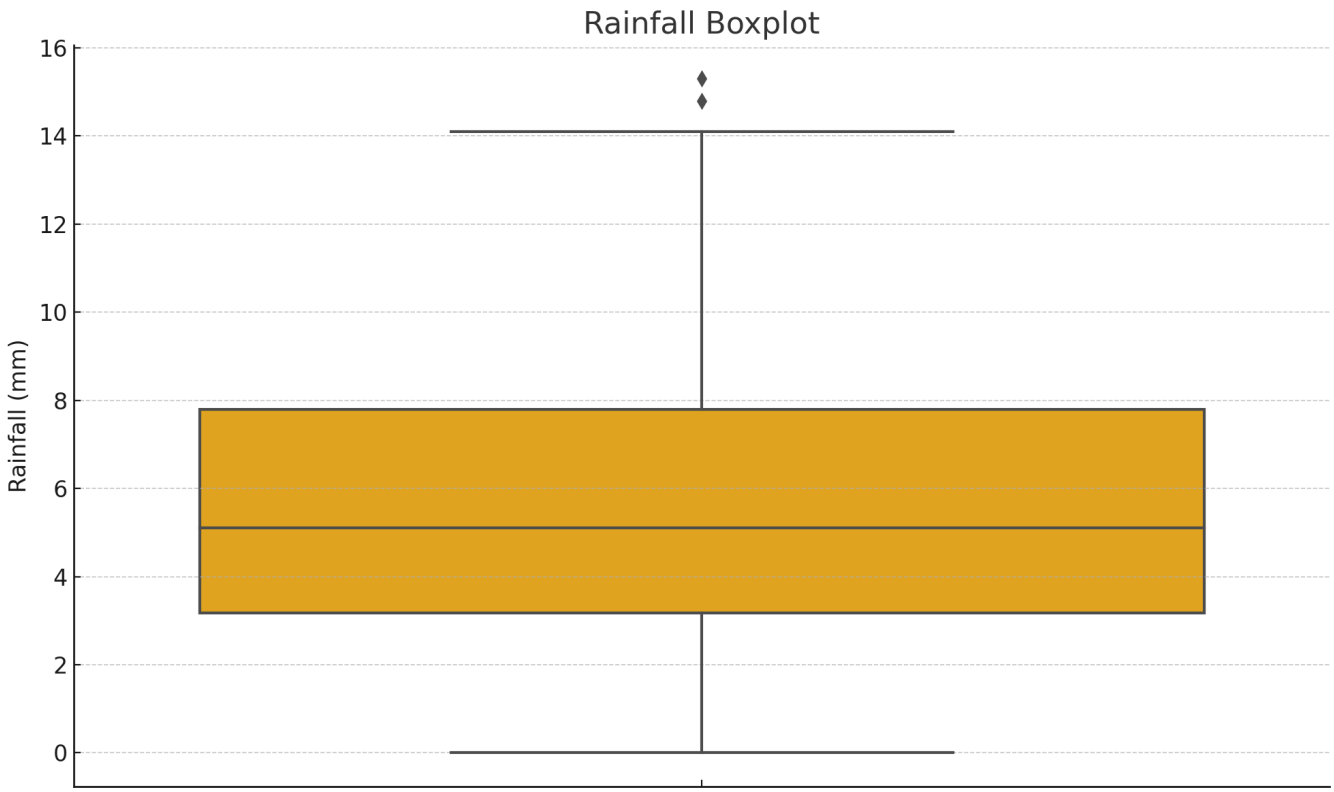
Temperature Trend Over Time



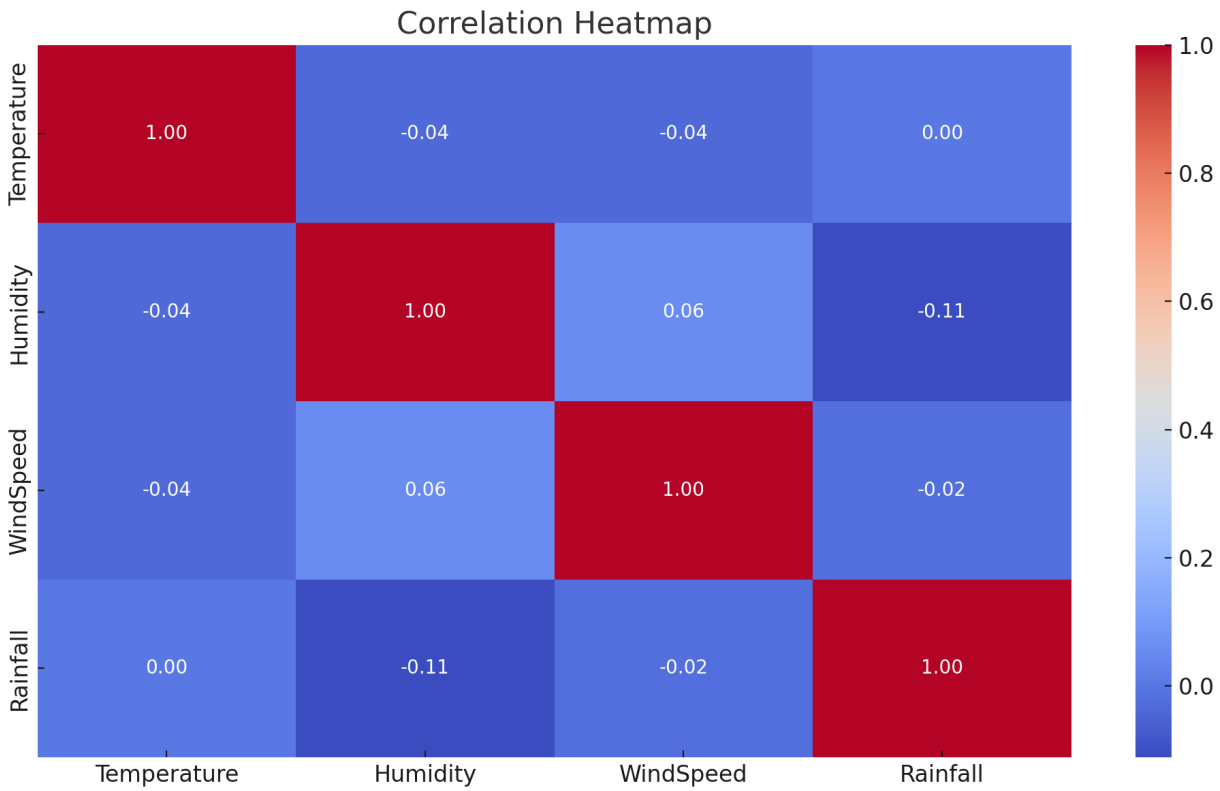
Humidity Distribution



Rainfall Boxplot



Correlation Heatmap



Conclusion

From the exploratory data analysis performed on the weather dataset, the following points are observed:

1. Temperature shows visible variation across the recorded period, indicating changing weather conditions.
2. Humidity values are centered around an average level with a reasonably normal spread.
3. Rainfall values show a wide range, which suggests the presence of both dry and wet days.
4. The correlation heatmap provides an overview of the linear relationships between the variables, which is useful for further predictive modeling.

Overall, EDA helps in understanding data behavior and supports better decision making.

References

1. Pandas Documentation: https://pandas.pydata.org
2. NumPy Documentation: https://numpy.org
3. Matplotlib Documentation: https://matplotlib.org
4. Seaborn Documentation: https://seaborn.pydata.org
5. Synthetic weather dataset generated using Python and NumPy.